Problem Set 6: Reactor Kinetics and Control

Due: Friday, February 27, 2026

Instructions

- Show all work, including the equations used and unit conversions.
- Assume the following data for a U-235 thermal reactor unless otherwise stated:
  - Effective Delayed Neutron Fraction: $\beta = 0.0065$
  - Decay Constant of Precursors (One-group average): $\lambda = 0.08\ \text{s}^{-1}$
  - Mean Generation Time: $l = 10^{-4}$ s

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1. **The "Bomb" vs. The Reactor (Prompt vs. Delayed Kinetics)**
   To appreciate why delayed neutrons are essential, let us compare two hypothetical scenarios where we insert the *exact same* amount of reactivity: $\rho = 0.002$ (0.2%).

   (a) **Scenario A (No Delayed Neutrons):** Imagine a fictional universe where all neutrons are prompt. The period is given by $T \approx l/\rho$. Calculate the Reactor Period T. Then, calculate the ratio of power at $t = 2$ seconds to initial power ($P(2s)/P_0$).

   (b) **Scenario B (Real World):** In our universe, we have delayed neutrons ($\beta = 0.0065$). Since $\rho < \beta$, we use the stable period approximation: $T \approx (\beta - \rho)/(\lambda\rho)$. Calculate the Reactor Period T. Then, calculate the power ratio $P(2s)/P_0$.

   (c) **Comparison:** Briefly comment on the difference. Which scenario is controllable by a human operator?

2. **Calculating Reactivity from Period**
   During a startup procedure, a reactor operator withdraws a control rod and observes that the reactor power is doubling exactly every **60 seconds** ($T_{doubling} = 60$ s).

   (a) Convert the "Doubling Time" to the **Reactor Period** (T). *(Hint: $P(t) = P_0 e^{t/T}$ vs $P(t) = P_0 2^{t/T_d}$).*

   (b) Using the one-group Inhour Equation (full form or appropriate approximation), calculate the Reactivity ρ inserted.

   (c) Convert this reactivity into units of **Dollars** (\$) and **Cents** (¢).

3. Overcoming the Power Defect

A PWR reactor is critical at Hot Zero Power (HZP) with a fuel temperature of 300°C. The operators wish to bring the reactor to Full Power (FP), where the average fuel temperature is 900°C.

The fuel temperature coefficient (Doppler coefficient) is $\alpha_{fuel} = -2.5 \times 10^{-5}\,(\Delta k/k)/°\text{C}$.

(a) Calculate the negative reactivity added by the fuel heating up (the "Doppler Defect").

(b) To stay critical at full power, the operator must withdraw control rods to add positive reactivity. If a typical control rod bank is worth 500 pcm (where 1 pcm = $10^{-5}\Delta k/k$), how many banks of rods must be withdrawn to compensate for this heating?

4. Naval Safety Analysis: The "Cold Water Slug"

One of the most dangerous transients for a naval reactor is the "Cold Water Accident." This occurs if a main coolant pump is started suddenly, sending a slug of inactive (cold) loop water into the hot core.

The Scenario:

- The reactor is operating at a stable low power with $T_{avg} = 300°\text{C}$.
- A slug of water at 150°C enters the core.
- The Moderator Temperature Coefficient is $\alpha_{mod} = -4.0 \times 10^{-4}\,(\Delta k/k)/°\text{C}$. (Note the negative sign).

Note for analysis: Assume a conservative "worst-case" scenario where the entire core is instantly filled with the 150°C water before the fuel heat capacity can significantly warm it up. We are interested in the initial reactivity insertion.

(a) Calculate the change in temperature ΔT.

(b) Calculate the resulting reactivity insertion $\rho = \alpha_{mod}\Delta T$. Be careful with signs!

(c) Compare this ρ to the delayed neutron fraction $\beta = 0.0065$. Is the reactor **Sub-critical**, **Delayed Super-critical**, or **Super-Prompt Critical**?

(d) If the reactor is Super-Prompt Critical, estimate the resulting period using the prompt approximation ($T \approx l/(\rho - \beta)$). Why is this situation catastrophic?

5. Decay Heat Calculations

Even after the chain reaction stops (scram), the fuel continues to generate heat due to the radioactive decay of fission products (betas and gammas). The Wigner-Way formula approximates the power $P(t)$ at time t after shutdown, following an operation time T_0 at power P_0:

$$\frac{P(t)}{P_0} = 0.066\left[t^{-0.2} - (t + T_0)^{-0.2}\right]$$

(where t and T_0 are in seconds).

(a) A reactor operates at 3000 MW$_{th}$ for 1 year (3.15×10^7 s) and then scrams. Calculate the thermal power generated **1 second** after shutdown.

(b) Calculate the power generated **1 hour** after shutdown.

(c) Why is this residual heat a major engineering challenge for accident scenarios (like Fukushima or TMI)?